

Hardness Sampling: Exploring Instance Hardness in Pool-based Active Learning

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Abstract. Predictive tasks usually require labeled data, which are often costly or impractical to acquire. Active Learning (AL) methods address this bottleneck, enabling the induction of predictive models with reduced label annotation cost. They optimize the labeling process by strategically sampling a subset of the unlabeled examples and sequentially selecting instances until a suitable labeled set is obtained. Several sampling strategies have been proposed in the literature. Although those based on Pool-based AL (PAL) scenario have presented some of the best results, they still have known limitations. This paper investigates how to address these limitations. PAL iteratively selects instances from a data pool to be labeled by an oracle and incorporated into the training set, increasing its representativeness for the current classification task. Its strategies, however, are limited by the active learners' bias within the exploration-exploitation tradeoff. For instance, non-agnostic strategies are biased toward prospective sampling, as they depend on the learner's assumptions. This has often resulted in suboptimal performance in the early stages of AL. To mitigate this issue, more exploratory biases have been introduced. Nevertheless, current strategies continue to exhibit shortcomings in data selection. On the other hand, recent studies have shown that identifying instances that are hard to classify significantly impacts the predictive performance of classifiers. This paper proposes a new non-agnostic PAL strategy, Hardness Sampling (HardS), which is based on Hardness Measures (HMs). HMs employ the Instance Hardness (IH) concept to identify instances with a higher probability of being misclassified. According to experiments carried out across diverse datasets, HardS is a competitive alternative to classical approaches, in particular Uncertainty Sampling, Expected Error Reduction, and Density-Weighted methods. The experimental results also suggest that some groups of HMs introduce a self-regulating balance between exploratory and prospective sampling biases, addressing such a key PAL challenge.

Keywords: Pool-based active learning · Instance hardness · Hardness measures · Hardness sampling · Query strategies.

1 Introduction

Instance labeling is increasingly important in Machine Learning (ML), as several applications can produce data, often in high volume and velocity. As a result, these data are usually unlabeled, since data labeling can be unfeasible or have a high cost, depending on the expertise needed. When feasible, data labeling can be carried out by people in ML applications and, depending on the complexity of the label process, high-skilled professionals in the data domain, known as oracles, may be needed [27]. Active Learning (AL) is an alternative to human labeling, reducing time, data needed, and cost. AL selectively queries the most informative data instances, mirroring human curiosity [27].

This study focuses on a specific AL approach, Pool-based AL (PAL) [18], due to its popularity in AL research [27]. PAL deals with a small labeled dataset $\mathcal{L}$ and a large pool of unlabeled data $\mathcal{U}$. A typical PAL process selects a subset of instances contained in $\mathcal{U}$ to be labeled by the oracle to increase the size of $\mathcal{L}$. The method used to select these instances relies on the query strategy applied, which defines the criteria for identifying whether an instance is critical for inducing a predictive model capable of generalizing beyond the available data. PAL strategies can be classified as *agnostic* or *non-agnostic* based on their dependency on predictions. Due to their reliance on the learner’s prediction, which is an estimation, non-agnostic strategies exhibit more prospective bias, i.e., they focus on the decision boundary, while agnostic strategies tend to be more exploratory [25]. Since combining these biases may lead to better results, efforts have been made to incorporate more exploratory biases in non-agnostic strategies [10, 13, 19, 32].

Another important concept that forms the basis for this work is Instance Hardness (IH). Specifically, in classification problems, IH can be understood as a property indicating the probability that an instance will be misclassified. However, IH has two key limitations: computational cost (requiring evaluation by multiple classifiers per instance) and explanatory limitations (it detects hardness but cannot explain it or generalize to other instances) [29]. With this in mind, Smith et al. (2014) propose Hardness Measures (HMs), which are intended to indicate the reason why an instance is harder to classify [29]. In addition, recent studies [1, 21] have also summarized HMs and introduced new ones.

Although HMs have been applied for different purposes in literature [17, 21, 23, 30, 31], their applicability within PAL remains unexplored to the extent of our knowledge. To address this gap, this work aims to investigate the use of HMs as informativeness measures in PAL. This exploration is conducted through the development of Hardness Sampling (HardS), a non-agnostic query strategy. The strategy selects instances based on their potential hardness – i.e., the value attributed by some HM to instances in $\mathcal{U}$ – offering an innovative approach to the field. The results demonstrate that HardS is competitive against other non-agnostic approaches. Moreover, they suggest that certain groups of measures can self-regulate the trade-off between exploratory and prospective biases during the sampling process.

2 Background and Related Work

This section briefly reviews the core concepts covered in this work. Section 2.1 provides a detailed description of the PAL scenario, distinguishing between agnostic and non-agnostic strategies concerning their sampling biases. It also examines relevant literature and recent advances in non-agnostic approaches. Section 2.2 formally introduces the concept of IH and presents HMs, while explaining the differences between various HM categories and their implementations.

2.1 Non-agnostic PAL and Query Strategy Bias

ML evolving predictive models can mimic the human curiosity of a student asking for directions or participating more broadly in the context of human active learning. They query the instances where they are least certain or, generally, query the instances that are more informative in a given dataset. In this way, they can establish a “line of inquiry” that can accelerate the generalization process for a particular problem [27] while avoiding the costs of unnecessary labels. In a pool-based scenario [18], an AL strategy must be able to select the most informative instance(s) from a pool of unlabeled data $\mathcal{U}$, so that the oracle can be queried about its true label. However, the approach to querying the oracle directly depends on the strategy employed to characterize an informative instance.

More formally, given the instance space $\mathcal{X}$ and the class set $\mathcal{Y}$, consider the dataset $\mathcal{D} = \{\langle \mathbf{x}_i, y_i \rangle \mid \mathbf{x}_i \in \mathcal{X} \wedge y_i \in \mathcal{Y}\}$, which maps instances in $\mathcal{X}$ to their respective classes in $\mathcal{Y}$. Additionally, let there be a pool of unlabeled instances $\mathcal{U} \subset \mathcal{X}$ and a labeled set of instances $\mathcal{L} \subset \mathcal{D}$. A generic query strategy for PAL can then be described by:

$$\mathbf{x}^* = \operatorname{argmax}_{\mathbf{x}_i \in \mathcal{U}} \phi(\mathbf{x}_i), \quad (1)$$

where $\mathbf{x}^*$ represents the most informative example in $\mathcal{U}$ based on some utility measure ϕ . This utility measure assigns a degree of informativeness to all instances in the pool. Therefore, $\mathbf{x}^*$ can be queried for label annotation, and then a new labeled set $\mathcal{L}' = \langle \mathbf{x}^*, y^* \rangle \cup \mathcal{L}$ is produced. Hence, the set of strategies dependent on ϕ is as diverse as there are different choices for ϕ .

PAL strategies can generally be classified as *agnostic* or *non-agnostic* based on their dependency on model predictions. Non-agnostic strategies rely on the active learner’s estimated decision boundary to assess instance informativeness, making them more *prospective* as they focus on promising regions of the input space. In contrast, agnostic strategies do not assume the correctness of this boundary, leading to a more *exploratory* approach that prioritizes instances from dense and unknown regions of the feature space [25]. As a result, non-agnostic strategies often prove more effective after the initial sampling stages, as the decision boundary is generally more reliable with a larger number of labeled instances, while typically underperforming agnostic strategies during initial iterations.

Uncertainty Sampling (US) [18], arguably the most widely used non-agnostic PAL strategy in practice [27], exemplifies a prospective behavior. Its core idea is to make the utility measure a function of the model’s confidence, so that the learner can avoid querying instances it is already confident about, allowing it to focus on the more challenging ones [27]. Another well-known strategy, Expected Error Reduction (EER) [26], focuses on selecting instances that, once labeled, have a higher probability of reducing the future error [27]. These strategies, however, are limited by their individual instance analysis, which risks selecting poor queries at the beginning of the learning process (e.g., outliers), limiting their potential [27].

To address this limitation, Settles and Craven [28] proposed Density-Weighted methods (DW) via the Information Density (ID) framework, which posits that informative instances should simultaneously exhibit high information content and representativeness of the underlying data distribution [27]. Formally, the utility $\phi_A(\mathbf{x}_i)$ of an instance $\mathbf{x}_i$ under strategy A is weighted by its average similarity to other unlabeled instances scaled by factor β , thereby increasing the prioritization of representative points in $\mathcal{U}$ and introducing an exploratory bias. The Training Utility (TU) function [11] extends this framework by further weighing instances inversely to their similarity to labeled data scaled by factor δ , creating a stronger exploratory bias that actively avoids previously sampled regions.

Additionally, several other non-agnostic strategies have been developed to incorporate the structure of unlabeled data. Adaptive AL [19] introduces a self-adjusting mechanism, while strategies such as Graph Density [10], Querying Informative and Representative Examples [14], and Representative Sampling [32] jointly consider both informativeness and representativeness. As an alternative approach, AL by Learning [13] dynamically selects the most appropriate strategy based on problem characteristics.

2.2 Hardness Measures

In their foundational work, Smith et al. (2014) defined Instance Hardness (IH) as the property describing how likely an instance $\mathbf{x}_i$ is to be misclassified by a generic learner [29]. While IH helps identify problematic instances, it does not inherently explain their hardness. To address this, the authors also proposed a comprehensive set of Hardness Measures (HMs) that evaluate how specific aspects may contribute to IH, enabling systematic analyses and processing of hard instances [29]. This framework was later expanded by Arruda et al. (2020), who adapted classification complexity measures to the IH domain [1, 20]. As shown in Table 1, these measures were recently synthesized into five distinct categories by Lorena et al. (2024) [21]. This section provides a concise overview of these categories and their constituent measures.

Neighborhood-based measures assume that instances surrounded by examples from other classes are harder to classify than those located in regions with a higher density of their own class [21]. Accordingly, the kDN measure quantifies $\mathbf{x}_i$ ’s hardness based on how many of its k-nearest neighbors belong to a different

Category	Measure	Acronym
Neighborhood	k-Disagreeing Neighbors [29]	kDN
	Frac. nearby instances different class at instance-level [1]	N1I
	Ratio of intra-extra class distances at instance level [1]	N2I
	Local set cardinality at instance-level [1]	LSCI
	Local set radius [1]	LSR
	Usefulness [1]	U
Likelihood	Harmfulness [1]	H
	Class Likelihood [29]	CL
Feature-based	Class Likelihood Difference [29]	CLD
	Frac. features in overlapping areas [1]	F1I
	Min. distance to overlapping areas of features [1]	F2I
	Mean distance to overlapping areas of features [1]	F3I
	Max. distance to overlapping areas of features [1]	F4I
Tree-based	Disjunct Size [29]	DS
	Disjunct Class Percentage [29]	DCP
	Tree Depth (pruned) [29]	TD _p
	Tree Depth (unpruned) [29]	TD _u
Class Balance	Class Balance [29]	CB
	Majority Value [29]	MV

Table 1. Hardness measures as categorized by Lorena et al. (2024) [21].

class [29]. N1I extends this idea by constructing a minimum spanning tree over the dataset, supposing that harder instances have more connections with other classes in the tree [1]. Meanwhile, N2I evaluates the ratio between the distance to the nearest neighbor of the same class and the distance to the nearest neighbor from an opposing class (nearest enemy), indicating that instances closer to opposing classes or farther from their own are harder [1]. The LSCI, LSR, and U measures rely on the local set concept – the set of instances closer to $\mathbf{x}_i$ than its nearest enemy [1]. While LSCI counts how many same-class instances belong to this set, LSR computes its radius, and U measures how many instances include $\mathbf{x}_i$ in their own local sets. Finally, the H measure assesses hardness based on how many instances have $\mathbf{x}_i$ as their nearest enemy [1].

Likelihood-based measures evaluate hardness by comparing an instance’s characteristics with its class patterns, based on the posterior probability $P(\mathbf{x}_i|y_i)$ [29]. This probability is computed similarly to the approach used in the Naive Bayes classification algorithm, where class likelihood is derived from independent feature probability distributions [21, 34]. The CL measure identifies harder instances as those with a low probability of belonging to their own class. Its extension CLD further considers instances harder when they both deviate from their class patterns and show significant similarity to other classes [21].

Feature-based measures try to quantify the fact that instances with feature values outside class overlapping regions are generally easier to classify [1, 21]. The F1I measure does this by calculating the fraction of features in overlapping

areas, while F2I, F3I, and F4I measures estimate the minimum, average, and maximum distances of instance $\mathbf{x}_i$ to the center of these regions.

Tree-based measures operate by first constructing a decision tree from the dataset, under the assumption that harder instances require more decision splits for correct classification [21, 29]. The DS measure estimates the number of splits based on the relative size of the classifying disjunct (leaf node), as smaller disjuncts suggest that more splits were needed for classification. Similarly, DCP employs a pruned tree and calculates the class percentage within the instance’s disjunct, where hardness correlates with minority class representation. Finally, TD assesses hardness by the relative depth of the classifying disjunct, with two variants (TD_p for pruned and TD_u for unpruned trees).

Class balance-based measures consider instances belonging to a minority class as harder to classify, as they are more susceptible to classification errors [21]. Both CB and MV measures [29] characterize the difficulty of an example based on its class distribution. The difference is that while CB takes into account all classes that make up the classification problem, MV considers only the proportion between the number of instances that have the same class as $\mathbf{x}_i$ and the number of instances belonging to the most representative class.

3 Hardness Sampling

In this section, Hardness Sampling (HardS) is proposed, a new non-agnostic query strategy designed to sample instances that are potentially hard to classify. This strategy aims to explore how HMs behave in PAL by employing a specific measure as the utility function. In Equation 1, this corresponds to setting $\phi(\mathbf{x}_i) = \text{Hard}(\mathbf{x}_i)$, where $\text{Hard}(\cdot)$ represents some HM. Notwithstanding, the measures presented in Section 2.2 rely on knowledge about the true label y_i of instance $\mathbf{x}_i$, which limits their applicability in PAL. To overcome this issue, the predictions made by the active learner were used as instance labels. Although this approach may be vulnerable to learner-induced bias, the model’s evolution over successive iterations is expected to attenuate it.

Another constraint arising from HardS is that HMs assume that the problem at hand involves more than one class. However, when it comes to the early stages of AL, the learner is prone to classify all unlabeled instances as belonging to the same class. To address this issue, random sampling (Rnd) was adopted as a fallback method for cases where the learner’s predictions prevent HMs from being extracted. Rnd was chosen because it relies solely on the random selection of unlabeled instances, thereby avoiding the introduction of additional bias into the process.

The strategy presented in this work fundamentally differs from most conventional non-agnostic approaches (Section 2.1) by analyzing model predictions through their complex interdependence rather than in isolation. Instead of using predicted label likelihoods directly, measures that integrate these predictions are computed, creating a more holistic assessment. In this sampling framework, an instance’s confidence becomes less critical than its potential hardness – the

estimated hardness for any model to classify it correctly, assuming the learner’s problem assumptions hold. As a result, HardS adapts sampling to the learner’s evolving problem perception, enabling autonomous transitioning between exploratory and prospective query biases, without requiring direct intervention.

4 Methodology and Experimental Configuration

For a compelling presentation of HardS as a new PAL strategy, it is crucial to benchmark it against some well-known approaches. To this end, a set of methods $\mathcal{Q}$ was gathered, comprising HardS alongside other representative PAL strategies. Each method in $\mathcal{Q}$ was combined with a fixed learning algorithm a and evaluated across a heterogeneous collection of datasets $\mathcal{S}$. This experimental design allows us to measure the performance of each strategy $q \in \mathcal{Q}$ on dataset $\mathcal{D} \in \mathcal{S}$, enabling direct comparisons across methods for individual datasets. These comparisons, in turn, provide insights into the overall competitiveness of HardS across the entire collection $\mathcal{S}$. The remainder of this section describes the full experimental configuration: the method set $\mathcal{Q}$, the learning algorithm a , and the dataset collection $\mathcal{S}$, followed by the methodology for benchmarking HardS against the other strategies.

The $\mathcal{Q}$ set was structured to include methods representing various strategies from PAL literature (classic methods), along with those introduced by the HardS. The classic methods chosen were: Margin Sampling (MS) [27], representing the US strategy; EER_{ent} [26], which implements EER by minimizing the entropy-based loss function; and ID and TU representing the DW strategy.

In this implementation, both ID and TU weighted MS with α and β set to 1, using the Euclidean distance to measure similarity between instances. Rnd was included as a baseline to assess whether any of the methods presented offered an advantage over the random selection of unlabeled instances. For HardS, all measures from Table 1 were selected. In the case of neighborhood-based HMs, the Gower distance [12] was used, with $k = 10$ for kDN.

We adopted a single classification algorithm, Support Vector Machine (SVM) [4], as it has been a baseline classifier in recent studies benchmarking PAL strategies [22, 33]. The hyperparameters were fixed at $C = 1$ and γ set to the inverse of the product between the number of features and the data variance.

Finally, the collection $\mathcal{S}$ has 80 classification datasets, from [16], selected based on those used by Pereira-Santos et al. (2019) [25]. These datasets were chosen to evaluate the elements of $\mathcal{Q}$ under various conditions, such as dataset size and class imbalance. The number of classes ranged from 2 to 30, with most datasets containing up to 5. Dataset sizes vary significantly, spanning from hundreds to tens of thousands of instances. The datasets contain between 2 and 167 attributes, most of which are numerical. To ensure compatibility with the chosen learning algorithm, numerical attributes were standardized, and one-hot encoding was applied to convert nominal attributes into numerical form.

To evaluate the performance of method q on dataset $\mathcal{D}$, $\mathcal{D}$ was split into $\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{test}}$, where the training set was used to simulate the PAL process,

while the test set served to evaluate the performance of the learner guided by q . To ensure a more reliable evaluation of each pair $(q, \mathcal{D})$, stratified 5-fold cross-validation was adopted, where one fold was used for testing and the remaining folds for training. Let $\mathcal{D}_{train}^k$ and $\mathcal{D}_{test}^k$ denote the training and test sets obtained from the k -th fold, respectively.

Next, the initial labeled and unlabeled sets assumed at the beginning of the PAL process were derived from the training set, so that $\mathcal{D}_{train}^k = \mathcal{L}_0^k \cup \mathcal{U}_0^k$. Additionally, the initial labeled set size was set to $|\mathcal{L}_0| = \max(5, c)$, where c denotes the number of classes in $\mathcal{D}_{train}^k$. This ensures that $\mathcal{L}_0$ contains at least one example from each class. If the number of classes was smaller than 5, the remaining instances were randomly selected to reach 5 labeled samples.

Given the initial conditions, the AL loop was carried out over 100 iterations for each fold, with one instance being labeled per iteration based on method q . For each iteration i and fold k , the performance of the model θ_i^k – trained on $\mathcal{L}_i^k$ – was evaluated on $\mathcal{D}_{test}^k$ using Cohen’s Kappa coefficient [3], denoted as κ_i^k . This measure was selected due to its inherent suitability for multiclass problems (particularly relevant for the chosen datasets) and its robust handling of class imbalance [6]. This coefficient is especially valuable as it explicitly accounts for random chance θ_i^k ’s predictions while treating true and predicted labels as distinct probability distributions [2].

To evaluate this pipeline, the performance of each method q on dataset $\mathcal{D}$ could be submitted to two complementary breakdowns: (i) iteration-specific performance, captured by the fold-averaged Kappa score of the i -th iteration, $\bar{\kappa}_i = \frac{1}{5} \sum_{k=1}^5 \kappa_i^k$; and (ii) overall performance, quantified via the Area Under the Learning Curve (AUC) [5] computed independently across each fold’s trajectory $(\kappa_0^k, \dots, \kappa_{100}^k)$ and then averaged across all 5 folds. These perspectives enable us to compare methods’ general performance on a given dataset and investigate whether their success is linked to specific stages of the PAL process.

Although AUC and $\bar{\kappa}_i$ are useful for assessing individual datasets, they are not ideal for evaluating overall performance across the entire collection $\mathcal{S}$. This is because comparisons of average values obtained by a method can be unreliable due to the varying difficulty levels presented by each dataset [25]. Hence, in addition to per-dataset AUC value comparisons, cross-dataset comparisons were conducted using mean ranking positions of $\bar{\kappa}_i$ across iterations for stage-specific evaluation and mean ranking positions of AUC values for overall performance, avoiding potentially biased direct value comparisons. For iteration-specific analysis, ranking curves [9] were employed to visualize performance trends and contrast exploratory/prospective biases across strategies. For overall performance evaluation, the Friedman test ($\alpha = 0.05$) was applied with χ_F^2 and F_F [15] distributions, followed by the Nemenyi post-hoc test ($\alpha = 0.05$) [8] to verify HardS’ statistical superiority.

The experiments were conducted on the Ubuntu 22.04 LTS GNU/Linux operating system, with kernel version 6.8.0-36-generic, using Python 3.8.19. Most of the classic methods in $\mathcal{Q}$ were implemented by the `modAL` [7] module, while additional extensions were developed to fully implement the TU and ID meth-

ods. The HardS strategy was applied using `modAL` in combination with the HMs’ implementation provided by the `PyHard` package [23]. The `scikit-learn` implementation of SVM was employed [24]. All experiments used the default parameters from the respective libraries. Datasets and code are available in the Git repository at <https://github.com/gscnogueira/Hardness-Sampling>.

5 Experimental Results

This section presents the key empirical findings from this study through two complementary analyses. Section 5.1 examines methods’ performance across iterations, revealing characteristic patterns in their AL behavior. Section 5.2 assesses how the observed behaviors affect HardS’ performance on individual datasets, comparing it to the top classical approach and the baseline method. Additionally, it evaluates whether there are statistically significant differences in the overall performance of methods. Together, these analyses offer a comprehensive evaluation of the proposed strategy.

5.1 Iteration-specific Performance

Figure 1 displays a graphical overview of the methods’ general behavior across the AL loop. Each curve represents the mean ranking position achieved by a method at a given query. HardS methods are represented by solid lines and markers, where shared marker shapes indicate methods using HMs from the same group. Moreover, Rnd is depicted by a thick red dashed line, while classical methods are shown with thick solid lines. A moving average was applied to all curves to improve visualization, with a sliding window width of 5 queries, balancing smoothness and detail preservation.

Both Rnd and most HardS methods in the upper half of the graph (above position 12) exhibited an initial advance in the mean ranking, followed by stabilization. An exception was kDN, which maintained continuous advance throughout the entire PAL process. In contrast, methods in the lower half generally displayed a decline – often after a brief advance – before stabilization. Among classical methods, distinct patterns emerged: TU and ID displayed an initial advance followed by a continuous decline, while MS and EER_{ent} experienced an abrupt decline before recovering in the latter stages.

A closer analysis of the upper half of the graph reveals that it is predominantly composed of HardS methods. Although TU initially performed on par with LSR and kDN, its curve showed a clear downward trend starting at query 10, while the others continued to improve in subsequent iterations. After that, MS was the only classical method to enter this region, becoming competitive with HardS only around query 50. This region featured exclusively neighborhood-, tree-, and likelihood-based HMs, suggesting that the criteria they adopt to identify hard instances are transferable to PAL.

The lower half of the graph accommodated most classical methods across the AL loop, often with mean ranking positions lower than those of HardS.

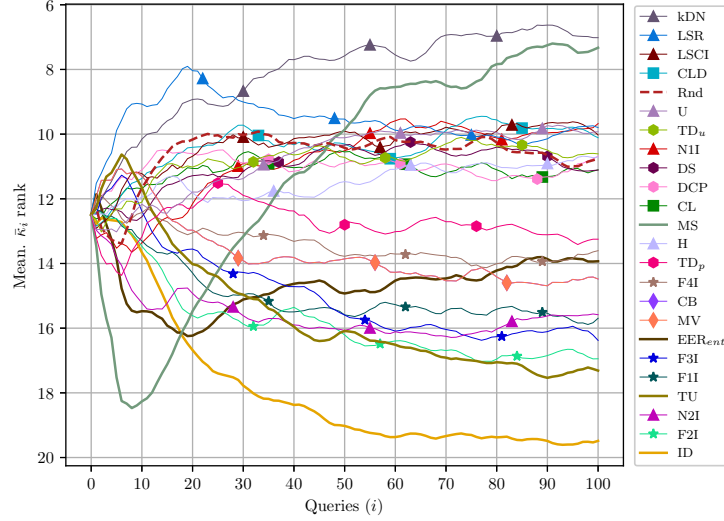


Fig. 1. Ranking curves for methods tested with the SVM algorithm. Curves with the same marker shape present HMs from the same group. Thick lines with no markers depict classical methods, and the red dashed line shows random sampling. The legend is sorted by the area under the ranking curve.

All feature- and class balance-based HMs remained in this region, with the two class balance measures showing overlapping curves — suggesting they are interchangeable as utility functions. Besides, while the neighbor-based measure N2I exhibited a downward trend similar to other methods in this region, the tree-based measure TD_p behaved differently, consistently maintaining superior positions compared to other curves in this sector.

The observed patterns suggest that methods derived from HMs from specific groups may offer enhanced capability to manage the exploration-exploitation trade-off in PAL. While exploratory-biased methods (ID, TU) showed an initial increase in their ranking positions, followed by a decline, prospective-biased methods (MS, EER_{ent}) exhibited the opposite behavior. Notably, most HardS methods derived from neighborhood, likelihood, and tree-based HMs demonstrated sustained advances in ranking positions or stability after initial queries, potentially indicating an adaptive capability. The sustained performance may reflect an emergent mechanism where increasing learner proficiency enables a more targeted instance selection. Such measures appear to naturally balance AL progression by avoiding premature prospective sampling, which — as demonstrated by MS and EER_{ent} — may compromise early-stage performance. Moreover, the distinct behavior exhibited by TD_p and N2I compared to their respective groups suggests that their specific implementation may hinder this adaptive mechanism from emerging.

5.2 Per-Dataset Performance

Table 2 presents a comprehensive evaluation of HardS’ effectiveness by comparing the performance of its best method against the top classical method and Rnd baseline. This comparison is based on the observed AUC gain across datasets, with negative values highlighted in red to indicate performance degradation.

	HardS*	Classic*	CG (%)	BG (%)		HardS*	Classic*	CG (%)	BG (%)
abalone-3class	kDN	EER _{ent}	13.67	11.45	pendigits	N1I	MS	7.60	10.26
artificial-charac...	CLD	MS	10.36	-1.24	phoneme	N2I	EER _{ent}	30.49	79.00
autoUniv-au7-700	N1I	EER _{ent}	24.65	23.14	pima-indians-diab...	DS	EER _{ent}	18.49	27.65
autoUniv-au7-cpd1...	CB	ID	2.92	111.92	qsar-biodegradation	kDN	EER _{ent}	30.50	11.30
balance-scale	CLD	MS	10.59	1.56	ringnorm	CB	TU	12.66	3.05
banana	DS	MS	22.73	9.87	robot-failure-lp5	F3I	EER _{ent}	1.07	-7.17
banknote-authenti...	LSR	MS	6.91	6.71	robot-nav-sensor-...	kDN	MS	5.16	12.56
bupa	kDN	ID	10.39	217.35	saheart	U	TU	51.49	103.37
car-evaluation	CLD	MS	34.69	63.05	seeds	U	TU	2.02	1.34
cardiotocography-...	TD _u	MS	26.42	22.02	spambase	LSR	EER _{ent}	49.35	7.40
connectionist-min...	CLD	MS	6.74	12.91	spect-heart	kDN	TU	12.57	20.58
connectionist-vowel	LSR	MS	19.79	20.33	statlog-australia...	LSCI	MS	16.82	15.17
ecoli	CL	MS	0.36	-0.64	statlog-german-cr...	kDN	TU	71.45	1099.76
flare	LSR	EER _{ent}	54.95	-11.24	statlog-heart	CLD	MS	6.74	-1.02
glass	LSCI	MS	12.72	13.05	statlog-image-seg...	LSR	MS	3.79	12.21
habermans-survival	TD _u	TU	-32.04	23.24	statlog-vehicle-s...	N1I	EER _{ent}	14.95	29.01
heart-disease-pro...	DS	MS	14.52	31.18	steel-plates-faults	CLD	MS	-0.37	-0.10
heart-disease-pro...	CLD	EER _{ent}	7.83	1.43	synthetic-control	N1I	MS	68.43	8.93
heart-disease-pro...	CB	TU	12.86	97.81	texture	LSR	MS	13.03	2.00
hepatitis	N2I	TU	-11.35	13.50	thyroid-ann	N1I	MS	5.31	825.36
horse-colic-surgical	DCP	EER _{ent}	19.28	14.47	thyroid-hypothyroid	DCP	MS	10.74	158.01
indian-liver-patient	H	ID	-46.30	1.77	thyroid-newthyroid	U	MS	8.12	13.72
ionosphere	LSR	MS	5.25	4.60	thyroid-sick-euth...	H	MS	63.89	1103.90
iris	kDN	MS	-0.55	1.71	tic-tac-toe	CLD	EER _{ent}	29.82	7.81
kr-vs-kp	TD _u	EER _{ent}	50.86	13.14	twonorm	N2I	TU	8.87	1.79
leaf	kDN	MS	-1.06	-0.35	user-knowledge	DS	MS	17.53	6.41
lymphography	U	EER _{ent}	21.58	14.44	vertebra-column-2c	CLD	MS	7.00	-6.86
magic	CLD	EER _{ent}	-22.74	-28.59	vertebra-column-3c	kDN	MS	9.74	14.31
mammographic-mass	DS	TU	9.82	11.96	volcanoes-a3	CLD	MS	10.66	397.91
mfeat-fourier	F4I	MS	44.91	1.79	volcanoes-b5	CB	MS	17.98	625.36
molecular-splice-...	TD _u	MS	64.17	268.19	volcanoes-e1	F4I	EER _{ent}	-28.36	566.26
monks1	kDN	MS	11.73	22.71	voting	H	TU	16.17	6.03
monks3	kDN	MS	8.70	21.00	waveform-v2	F2I	MS	21.84	10.98
movement-libras	kDN	MS	5.80	-0.51	wdbc	U	TU	19.73	10.92
mushroom	TD _u	MS	5.78	10.20	wholesale-channel	kDN	MS	7.37	9.05
musk	LSR	TU	191.05	60.99	wilt	F3I	MS	19.60	43487.19
nursery	H	MS	47.84	9.01	wine	F3I	TU	8.69	2.48
optdigits	LSCI	MS	15.50	13.32	wine-quality-red	CLD	MS	14.25	5.54
page-blocks	LSR	MS	3.58	218.98	wine-quality-whit...	F4I	MS	40.58	25.77
parkinsons	CLD	MS	5.05	8.90	yeast-4class	F4I	EER _{ent}	19.56	1.22

Table 2. Performance comparison of HardS across 80 datasets. Columns show: (1) Dataset name; (2) HardS* - best HardS method by AUC; (3) Classic* - best classical method by AUC; (4) Classic Gain (CG) - HardS*’s improvement over Classic*; (5) Baseline Gain (BG) - HardS*’s improvement over Rnd baseline. All AUC scores were computed using 5-fold cross-validation. Negative gains are highlighted in red.

The results show HardS’s superior performance, outperforming classical methods in 72 datasets, with an average gain of 17% (ranging up to 191%). While Rnd achieved marginally better results in 10 datasets, the advantage was minimal (typically < 1.5%). The table also highlights kDN, CLD, and LSR as the top-performing HardS methods, aligning with the trends shown in Figure 1.

These findings were further supported by statistical analysis. The Friedman test on AUC ranking positions yielded $p \approx 0$ for both χ_F^2 and F_F distributions, demonstrating significant differences in overall method performance. Post-hoc Nemenyi analysis delineated distinct performance ties with critical distance $CD \approx 1.82$. Figure 2 presents the mean ranking positions of each method in a critical distance diagram, grouping methods that share statistically equivalent positions.

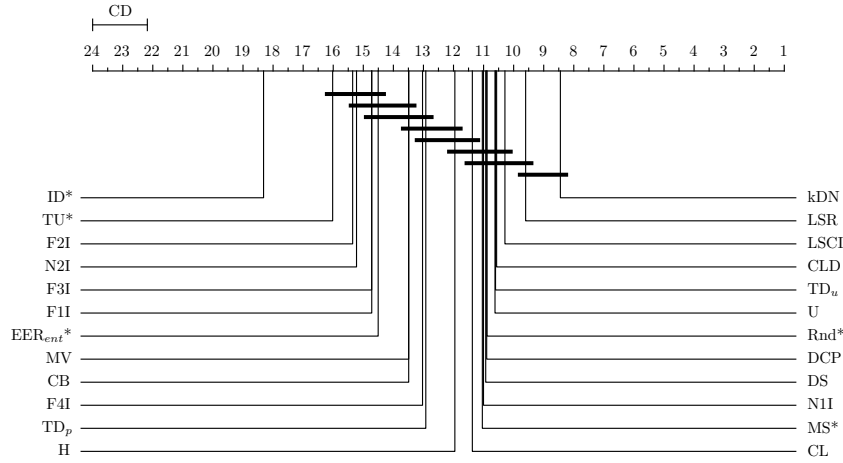


Fig. 2. Critical difference diagram for the Nemenyi post-hoc test on mean AUC of ranking positions ($\alpha = 0.05$, $CD \approx 1.82$). Groups connected by horizontal bars indicate no statistically significant difference. Classic methods and Rnd are marked with *.

The critical distance diagram demonstrates that every HardS method was statistically superior to at least one classical approach, with most outperforming both TU and ID. Among classical methods, only MS showed significant advantages over HardS, and solely against the lowest-performing HMs (feature-based, class balance-based, N2I, and TD_p). Notably, kDN was the only tested method to statistically outperform Rnd. This result may stem from dataset diversity, the constrained query budget assumed, or a combination of both factors. Nevertheless, these collective findings support HMs’ viability for PAL, performing comparably or better than established methods while maintaining consistent performance across varied conditions.

6 Conclusion and Further Work

This study presents a comprehensive investigation of HMs as informativeness criteria for non-agnostic PAL. The paper systematically evaluates their potential advantages over conventional strategies while examining their capacity to

balance exploration-exploitation trade-offs. Following a thorough review of classical PAL strategies were revisited and HMs surveyed, leading to the proposal of HardS. Through extensive experiments across multiple datasets, the proposed strategy was evaluated against classical methods and random sampling, assessing its comparative performance.

While only one HardS method (kDN) demonstrated statistically significant superiority over random sampling, the overall results establish HardS as a competitive alternative to conventional approaches. Notably, the analysis revealed advantages for neighborhood, tree, and likelihood-based HMs over both classic methods and other HM types (feature- and class balance-based measures). This divergent performance appears to stem from their ability to self-regulate the exploration-exploitation trade-off during the active learning process. However, the analysis also revealed that certain measures (N2I and TDu) may disrupt this mechanism due to their implementation of the hardness criteria.

Moreover, these findings suggest that neighborhood-, tree-, and likelihood-based hardness measures (excluding TD_p and N2I) represent the most suitable candidates for PAL applications. While not achieving statistical significance, the consistent performance patterns of kDN, LSR, and CLD across ranking curves and dataset evaluation methods provide empirical support for their superior regulation of the exploration-exploitation balance. Three key research directions are recommended for future work: (i) comparative evaluation of HardS against a wider array of PAL query strategies, (ii) comprehensive assessment across more diverse datasets, machine learning algorithms, and performance measures, and (iii) focused investigation of the bias-balancing mechanisms through controlled experiments. Such studies would not only validate the current findings but also provide deeper insights about HardS' potential.

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